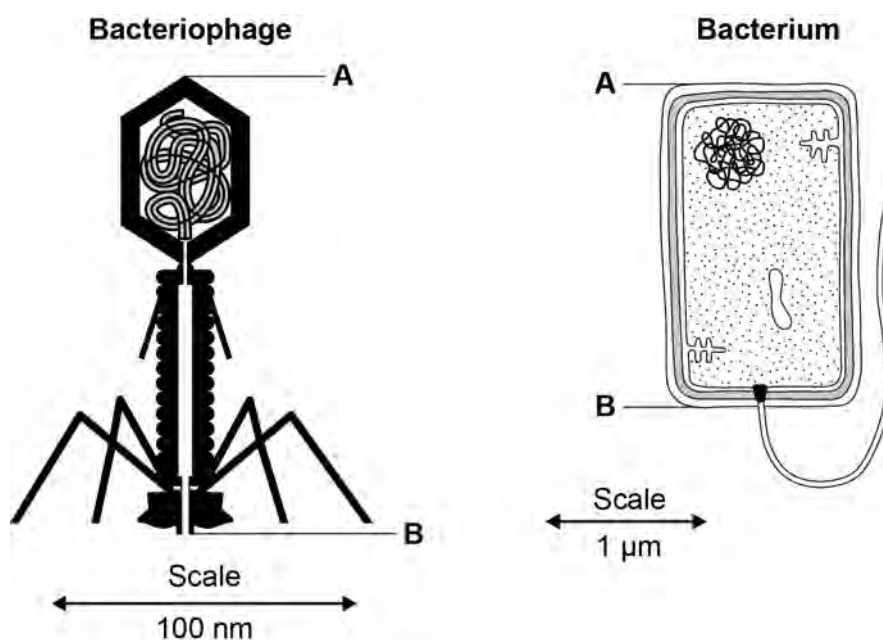


0 2

Bacteriophages are viruses that kill bacteria.

Figure 2 shows drawings of a bacteriophage and a bacterium.

Figure 2



0 2 . 1

Using **Figure 2** and your own knowledge, put a tick (✓) in the box next to the **only** correct statement about the structures of the bacteriophage and the bacterium.

[1 mark]

Both have ribosomes.

☐

Both have a cell-surface membrane.

☐

The bacteriophage has a capsid and the bacterium has a cell-surface membrane.

☐

The bacteriophage has a cell wall and the bacterium has a capsid.

☐

0	2	.	2
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Using the scales in **Figure 2**, calculate how many times longer the bacterium is than the bacteriophage.

Use the distance between the points labelled **A** and **B** on each drawing in your calculations. Show your working.

[2 marks]

The bacterium is _____ times longer

Question 2 continues on the next page

0 2 . 3

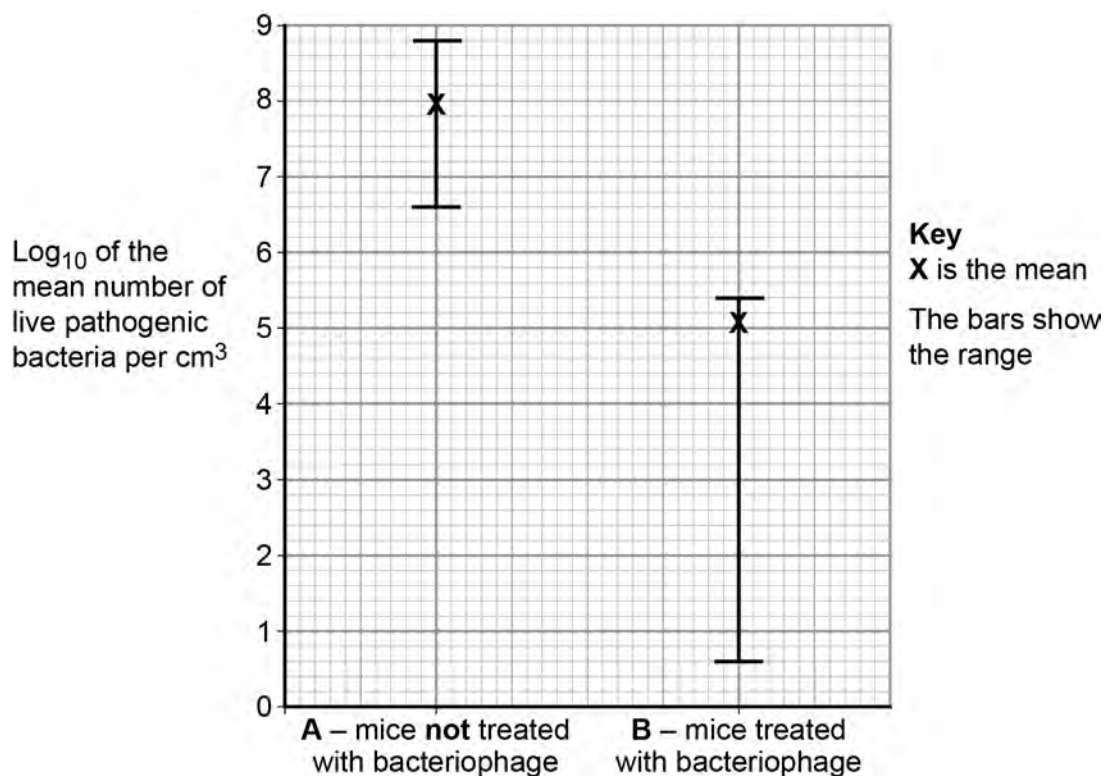
Scientists investigated the use of bacteriophages to treat lung infections caused by bacteria. They infected the lungs of mice with a pathogenic species of bacterium. The mice were then divided into two groups, **A** and **B**.

- The mice in group **A** were **not** treated with bacteriophage.
- The mice in group **B** were treated by breathing in a spray containing bacteriophage particles.

After 3 days, the scientists killed the mice and removed their lungs. They washed out each set of lungs with a set volume of liquid. The scientists determined the number of live bacteria in the liquid.

Figure 3 shows the scientists' results. **Figure 3** shows the mean and the range of the data about the mean for each group. Standard deviations of the means are **not** shown.

Figure 3



What would the scientists' null hypothesis be for this investigation?

[1 mark]

0 2 . 4

With some samples, the scientists decided they needed to carry out a series of dilutions of the sample before counting the bacteria.

Use evidence from **Figure 3** to explain why dilutions were necessary for some samples but not for others.

[2 marks]

0 2 . 5

Using only **Figure 3**, what can you conclude from these data about the effectiveness of the bacteriophage in treating this lung infection in mice?

Do **not** consider statistical analyses in your answer.

[3 marks]

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